

Notes: Bartelsman, Haltiwanger, and Scarpetta (AER, 2013)

Cross-Country Differences in Productivity: The Role of Allocation and Selection

Contents

1	Big picture	3
2	Data and measurement (key objects)	3
2.1	Harmonized firm-level cross-country database	3
2.2	Measured firm-level productivity objects	4
2.3	The three empirical moments the paper targets	4
2.4	Key empirical facts from the cross-country data	4
2.5	Interpretation of the measured productivity moments	5
3	Models and empirical specifications (all equations + interpretation)	5
3.1	Firm production technology	5
3.2	Demand structure and profits	5
3.3	Entry, survival, and selection	6
3.4	Input choice	6
3.5	Allocation versus selection in a simplified version	7
3.6	Benchmark calibration to the United States	7
3.7	Distortion experiments	8
3.8	Matching cross-country moments	9
4	Identification and instruments (what makes the quantitative exercise credible)	9
4.1	Why standard dispersion measures are not enough	9
4.2	Why the OP covariance is the preferred moment	10
4.3	What in the model identifies distortions	10
4.4	Why overhead labor and selection are indispensable	10
4.5	Relation to the literature	10

5	Tables and figures	11
5.1	Table 1: Within-industry productivity dispersion and OP covariance term	11
5.2	Table 2: Changes in productivity dispersion and OP covariance term	11
5.3	Figure 1: The impact of overhead costs on key moments	11
5.4	Figure 2: Distortion dispersion in the uncorrelated case	12
5.5	Figure 3: Distortion dispersion in the positively correlated case	12
5.6	Table 3: Matching the covariance moments country by country	12
5.7	Figure 4: Low-overhead robustness	13
5.8	Figure 5: Measurement error	14
5.9	Figure 6: Dispersion in the labor elasticity	14
6	Short summary	14
7	Conclusion	15
7.1	Core conclusion (what the paper establishes)	15
7.2	Economic intuition	15
7.3	Takeaway	15

1 Big picture

- **Question.** How much of cross-country productivity differences can be understood as misallocation generated by firm-level distortions, and what empirical moment is most informative about that misallocation?
- **Setting.** The paper combines harmonized firm-level evidence from the United States, Western Europe, and Central and Eastern Europe with a heterogeneous-firm model that features idiosyncratic distortions, endogenous selection, overhead labor, and quasi-fixed capital.
- **Core challenge.** Simple dispersion measures of firm productivity do not map cleanly into misallocation. The paper argues that the *covariance between firm size and productivity* is a more robust and revealing measure than dispersion alone.
- **Main empirical result.** The within-industry covariance between size and productivity is high in the United States, lower in Western Europe, and close to zero or even negative in some transition economies early in the sample. It rises substantially in Central and Eastern Europe during the 1990s.
- **Main theoretical result.** A model with heterogeneous firms, overhead labor, quasi-fixed capital, and idiosyncratic distortions can rationalize why labor-productivity dispersion exceeds TFP dispersion and why distortions weaken the size-productivity relationship through both **allocation** and **selection**.
- **Main quantitative result.** Calibrating the model to match the covariance between size and productivity implies sizable aggregate consequences. Countries with weaker covariance moments are interpreted as having larger distortions and lower implied consumption per capita.
- **Why this paper matters.** It shifts attention from productivity dispersion alone toward the size-productivity covariance, highlights the importance of endogenous selection in misallocation analysis, and offers a benchmark framework for thinking about how firm-level distortions map into aggregate performance.

2 Data and measurement (key objects)

2.1 Harmonized firm-level cross-country database

- **Coverage.** The paper uses harmonized industry-level moments constructed from firm-level manufacturing data for five industrial economies and three transition economies: the United States, United Kingdom, Germany, France, Netherlands, Hungary, Romania, and Slovenia.
- **Underlying sources.** The moments are built from business registers, social security records, corporate tax rolls, and enterprise surveys, using a “distributed micro-data” methodology with common code across countries.
- **Unit behind the moments.** The underlying data are firm level, not establishment level.
- **Industry aggregation.** To make countries comparable, manufacturing averages are constructed by aggregating industry-level moments using time-invariant US industry labor shares.

2.2 Measured firm-level productivity objects

- **Gross output.** Measured from firm sales deflated with industry-level output deflators.
- **Labor input.** Number of employees.
- **Capital input.** Book-value-based capital measure with adjustments described in the earlier Bartelsman–Haltiwanger–Scarpetta data paper.
- **Materials input.** Nominal materials expenditures deflated by industry-level materials deflators.
- **Revenue labor productivity.** The paper denotes this by LPR : revenue per worker using industry deflators.
- **Revenue total factor productivity.** The paper denotes this by $TFPR$: TFP based on nominal revenue deflated with industry-level deflators.
- **Physical productivity in the model.** The theoretical model distinguishes $TFPQ$, which is the physical productivity object driving firm heterogeneity.

2.3 The three empirical moments the paper targets

The paper focuses on three harmonized within-industry moments:

- the standard deviation of log revenue labor productivity, $STD(LPR)$,
- the standard deviation of log revenue total factor productivity, $STD(TFPR)$,
- and the Olley–Pakes covariance term between size and productivity.

The Olley–Pakes decomposition is

$$\Omega_t = \sum_i \theta_{it} \omega_{it} = \bar{\omega}_t + \sum_i (\theta_{it} - \bar{\theta}_t)(\omega_{it} - \bar{\omega}_t),$$

where ω_{it} is firm productivity and θ_{it} is the firm’s activity share in the industry.

- **Interpretation.** The second term is the OP covariance. It is positive when larger firms are systematically more productive than smaller firms.
- **Implementation in the data.** The paper uses log labor productivity for ω_{it} and labor shares for θ_{it} .
- **Why it matters.** The covariance term is interpreted as a direct summary of allocative efficiency: if productive firms are larger, resources are better aligned with productivity.

2.4 Key empirical facts from the cross-country data

- **Dispersion fact.** Within-industry labor-productivity dispersion exceeds within-industry TFP dispersion in almost all country-year-industry cells.
- **Covariance fact.** The OP covariance term is about 0.51 in US manufacturing, around 0.15–0.30 in Western Europe, and close to zero or negative in some transition economies at the start of transition.

- **Time-series fact.** The covariance term rises substantially in Hungary, Romania, and Slovenia between the early and late 1990s, while productivity dispersion changes much less.

2.5 Interpretation of the measured productivity moments

- **Revenue-based measures.** LPR and $TFPR$ are revenue-based productivity measures and therefore incorporate within-industry price variation.
- **Why the authors still use them.** These measures are widely available and, in the model, they remain positively correlated with physical productivity $TFPQ$.
- **Measurement issue.** The paper explicitly argues that some dispersion patterns may be less systematic than the covariance term partly because dispersion moments are more sensitive to measurement error.

3 Models and empirical specifications (all equations + interpretation)

3.1 Firm production technology

The production function is

$$Y_{it} = A_i \varepsilon_{it} (n_{it} - f)^{\gamma - \alpha} k_{it}^\alpha, \quad \gamma < 1.$$

- A_i is the permanent, firm-specific productivity component.
- ε_{it} is a transitory productivity shock observed after capital is chosen.
- n_{it} is labor, k_{it} is capital, and f is overhead labor.
- $\gamma < 1$ gives decreasing returns to scale, which helps prevent the most productive firm from taking over the entire market.

Interpretation.

- Overhead labor is the key friction that allows labor productivity to vary across firms even without distortions.
- Quasi-fixed capital means transitory shocks are mainly absorbed by labor, which generates heterogeneity in labor productivity and in the marginal revenue product of capital.

3.2 Demand structure and profits

The final good is a CES aggregator of firm outputs:

$$Y_t = N_t^{(\rho-1)/\rho} \left(\sum_i Y_{it}^\rho \right)^{1/\rho}, \quad \rho < 1.$$

This implies inverse demand

$$P_{it} = P_t \left(\frac{\bar{Y}_t}{Y_{it}} \right)^{1-\rho}, \quad \bar{Y}_t \equiv \frac{Y_t}{N_t}.$$

Firm profits are

$$\pi_{it} = (1 - \tau_i - \kappa_{it}) P_t \bar{Y}_t^{1-\rho} [A_i \varepsilon_{it} (n_{it} - f)^{\gamma-\alpha} k_{it}^\alpha]^\rho - w_t n_{it} - R_t k_{it},$$

where τ_i is a permanent idiosyncratic distortion and κ_{it} is a transitory distortion.

- **Interpretation of distortions.** They are scale distortions: anything that changes the effective return to producing or operating, such as taxes, regulation, financing favoritism, or uneven enforcement.
- **Markup channel.** Because $\rho < 1$, firms have market power. So even the “nondistorted” economy is not first-best.

3.3 Entry, survival, and selection

Ex post firm value is

$$W(A_i, \tau_i) = \frac{\mathbb{E}[\pi(A_i, \tau_i, \kappa_{it}, \varepsilon_{it})]}{1 - \chi}, \quad \chi = \frac{1 - \lambda}{1 + r},$$

where λ is the exogenous exit probability and r is the interest rate.

Free entry implies

$$W_e = \int \max\{0, W(A, \tau)\} dG(A, \tau) - c_e = 0,$$

where c_e is the sunk entry cost.

- **Interpretation.** Firms pay an entry cost before knowing their permanent productivity and distortion draws.
- **Selection channel.** Some firms exit immediately after learning their draws because expected operating profits are negative.
- **Why this matters.** Distortions affect not only how resources are allocated among operating firms, but also which firms survive in the first place.

3.4 Input choice

The labor first-order condition is

$$(\gamma - \alpha)\rho(1 - \tau_i - \kappa_{it}) P_t \bar{Y}_t^{1-\rho} (A_i \varepsilon_{it} k_{it}^\alpha)^\rho (n_{it} - f)^{(\gamma-\alpha)\rho-1} = w_t.$$

The capital first-order condition is

$$k_{it} = \left[\frac{\alpha \rho A_i^\rho P_t \mathbb{E} \left(\bar{Y}_t^{1-\rho} (1 - \tau_i - \kappa_{it}) (\varepsilon_{it} (n_{it} - f)^{\gamma-\alpha})^\rho \right)}{R_t} \right]^{1/(1-\alpha\rho)}.$$

- **Interpretation.** Labor responds to transitory shocks inside the period, while capital is chosen before those shocks are realized.
- **Economic role of quasi-fixed capital.** Capital does not adjust one-for-one to transitory productivity, so labor absorbs short-run fluctuations and labor productivity becomes dispersed even absent distortions.

3.5 Allocation versus selection in a simplified version

To build intuition, the paper studies a simplified version without transitory shocks and defines the “profit shock”

$$S_i = (1 - \tau_i)A_i^\rho.$$

In that case, optimal inputs satisfy

$$k_i^* = S_i^{1/(1-\rho\gamma)} K_x(w, \bar{Y}),$$

$$(n_i^* - f) = S_i^{1/(1-\rho\gamma)} \frac{R}{w} \frac{\gamma - \alpha}{\alpha} K_x(w, \bar{Y}).$$

- **Interpretation.** Inputs are increasing in the profit shock. Absent distortions, more productive firms are larger.
- **Canonical implication.** This generates a positive covariance between size and productivity in the absence of distortions.

Revenue per effective unit of labor is constant absent distortions, but labor productivity is not:

$$LPR_i = \frac{P_i Y_i}{n_i^*} = \frac{1}{1 - \tau_i} C - \frac{Cf}{(1 - \tau_i)n_i^*}.$$

- **Interpretation.** Overhead labor drives a wedge between marginal and average products of labor.
- **Key implication.** Even with no distortions, larger firms have higher labor productivity, so the OP covariance using labor productivity and employment can be positive.

There is also an endogenous exit threshold S_ℓ , below which firms do not operate. The threshold depends on wages, output, overhead labor, and technology parameters.

- **Selection implication.** Distortions worsen selection because some high-productivity firms with bad distortion draws exit, while some low-productivity firms with favorable draws survive.

3.6 Benchmark calibration to the United States

The benchmark “nondistorted” economy is calibrated to US manufacturing moments.

Key parameters are:

- $\gamma = 0.95$ (modest decreasing returns),
- $\alpha = 0.3$, implying a labor elasticity of 0.65,

- $\lambda = 0.10$,
- $r = 0.02$,
- $\delta = 0.10$,
- $\rho = 0.8$, implying a markup of 25%.

The remaining parameters—overhead labor, entry cost, and shock variances—are chosen to match US moments.

- The benchmark exactly matches a five-year entrant survival rate of 55%.
- It exactly matches the US OP covariance term of 0.51.
- It implies $\text{STD}(TFPR) = 0.47$ versus 0.39 in the data.
- It implies $\text{STD}(LPR) = 0.75$ versus 0.58 in the data.

Other benchmark implications:

- the OP covariance for TFPR is 0.40,
- the OP covariance for TFPQ is 0.91,
- overhead labor accounts for about 14% of total employment,
- entry costs absorb about 15% of output in the nondistorted steady state.

Interpretation.

- Overhead labor is central for matching the data pattern that labor-productivity dispersion exceeds TFPR dispersion.
- It is also central for matching the positive OP covariance and the pace of entrant failure.

3.7 Distortion experiments

The paper then varies the distribution of idiosyncratic distortions in two ways.

Uncorrelated distortions. Increasing distortion dispersion causes

- modest, nonmonotonic changes in $\text{STD}(LPR)$ and $\text{STD}(TFPR)$,
- declines in the OP covariance terms,
- lower survival,
- higher entry-cost shares,
- a higher capital-labor ratio,
- and lower consumption.

Positively correlated distortions and productivity. When the distortion is positively correlated with productivity, the quantitative effects are much stronger.

- TFP dispersion increases more.
- LPR dispersion first falls and then rises.
- The OP covariance terms for TFPQ, TFPR, and LPR fall sharply.
- Survival falls and consumption declines.

Interpretation.

- Uncorrelated distortions add “noise” to firm outcomes.
- Correlated distortions are worse because they actively push productive firms to be too small and unproductive firms to be too large or to survive too long.

3.8 Matching cross-country moments

The authors then ask whether cross-country differences in the data can be matched by changing only the distribution of distortions.

- They can exactly match the OP covariance term for every country in table 1.
- They cannot match cross-country dispersion moments nearly as well.
- Matching the covariance moments implies sizable consumption gaps relative to the United States.

Table 3 reports the implied model moments when the model is calibrated to match each country’s OP covariance. For example:

- the United Kingdom has data covariance 0.15 and implied consumption index 0.93,
- the Netherlands has covariance 0.30 and implied consumption index 0.97,
- Romania has covariance -0.03 and implied consumption index 0.88,
- Slovenia has covariance 0.04 and implied consumption index 0.89.

Interpretation.

- The covariance moment is the one the model organizes best.
- The data are telling the paper that the size-productivity link is more systematic across countries than productivity dispersion itself.

4 Identification and instruments (what makes the quantitative exercise credible)

4.1 Why standard dispersion measures are not enough

- A large literature uses dispersion in marginal products or revenue productivity as a direct sign of misallocation.

- This paper argues that dispersion moments alone are not enough, because they are affected by frictions that can exist even in a nondistorted economy, especially overhead labor and quasi-fixed capital.
- In particular, the empirical fact that $\text{STD}(LPR) > \text{STD}(TFPR)$ is impossible to reconcile with many standard models that equate average and marginal products of labor absent distortions.

4.2 Why the OP covariance is the preferred moment

- The size-productivity covariance is directly tied to the canonical prediction of heterogeneous-firm models: absent distortions, more productive firms should be larger.
- Distortions weaken that relationship, so the covariance provides a direct empirical handle on misallocation.
- The paper also argues that the covariance moment appears more robust to measurement error than dispersion moments.

4.3 What in the model identifies distortions

- The benchmark calibration fixes the technology, shock distributions, and curvature parameters using US moments and outside evidence.
- Cross-country differences are then attributed to differences in the distribution of distortions alone.
- The key idea is indirect inference in spirit: the model is disciplined by survival, dispersion, and covariance moments, and the distortion distribution is chosen to match the covariance moments across countries.

4.4 Why overhead labor and selection are indispensable

- Overhead labor makes labor productivity disperse even when distortions are absent.
- Endogenous selection means distortions affect not only operating decisions but also entry and survival.
- This is one of the paper's main methodological departures from the recent misallocation literature, which mostly abstracts from survival information in calibration.

4.5 Relation to the literature

- Relative to Hsieh and Klenow (2009), the paper emphasizes the size-productivity covariance rather than TFPR dispersion as the most robust diagnostic moment.
- Relative to Restuccia and Rogerson (2008), it adds explicit attention to endogenous selection and to the empirical fact that labor-productivity dispersion exceeds TFPR dispersion.
- Relative to Olley and Pakes (1996), it takes the covariance term seriously as a summary statistic for allocative efficiency and generalizes that logic across countries.

5 Tables and figures

5.1 Table 1: Within-industry productivity dispersion and OP covariance term

Read:

- The United States has the strongest size-productivity relationship: $OP = 0.51$.
- Western Europe is lower: United Kingdom 0.15, France 0.24, Netherlands 0.30, Germany 0.28.
- Transition economies are much lower: Hungary 0.16, Romania -0.03 , Slovenia 0.04.
- Labor-productivity dispersion exceeds TFPR dispersion everywhere. For example, the United States has 0.58 versus 0.39, and Romania has 1.05 versus 0.55.
- The table is the main motivation for the paper: the covariance term is highly systematic across countries, while dispersion is large everywhere.

TABLE 1—WITHIN-INDUSTRY PRODUCTIVITY DISPERSION AND OP COVARIANCE TERM
(Weighted averages of industry-level data, US industry weights)

	STD in revenue labor productivity	STD in revenue total factor productivity	OP covariance term
United States	0.58	0.39	0.51
United Kingdom	0.59	0.42	0.15
Germany	0.71	NA	0.28
France	0.53	0.23	0.24
Netherlands	0.55	0.15	0.30
Hungary	1.04	0.92	0.16
Romania	1.05	0.55	-0.03
Slovenia	0.80	0.22	0.04

Notes: Averages over 1993–2001 data. Industry-level firm based TFP measures not available for Germany.

Source: Firm-level database; see Bartelsman, Haltiwanger, and Scarpetta (2009).

TABLE 2—CHANGES IN PRODUCTIVITY DISPERSION AND OP COVARIANCE TERM
(Weighted averages of industry-level data, US industry weights)

	STD in revenue labor productivity	STD in revenue total factor productivity	OP covariance term
United States	0.02	0.00	0.09
United Kingdom	0.04	0.03	0.06
Germany	0.06	NA	0.14
France	NA	NA	NA
Netherlands	0.01	0.00	0.11
Hungary	-0.02	-0.03	0.18
Romania	0.03	-0.03	0.25
Slovenia	-0.06	-0.02	0.16

Notes: Change is difference in moments between the average value in 1997–2001 and the average value in 1993–1996. Data for France available only from 1992 to 1995 and for the United States for 1992 and 1997.

Source: Firm-level database; see Bartelsman, Haltiwanger, and Scarpetta (2009).

5.2 Table 2: Changes in productivity dispersion and OP covariance term

Read:

- The OP covariance rises strongly in the transition economies: Hungary $+0.18$, Romania $+0.25$, Slovenia $+0.16$.
- Western Europe and the United States see much smaller changes: United States $+0.09$, United Kingdom $+0.06$, Netherlands $+0.11$.
- Dispersion moments are comparatively stable over time.
- This is important because it suggests improvement in allocative efficiency during transition shows up more clearly in the covariance than in the dispersion moments.

5.3 Figure 1: The impact of overhead costs on key moments

Read:

- Raising overhead labor increases the dispersion of labor productivity and initially raises the covariance between size and productivity.
- But the covariance eventually flattens and then falls because higher overhead costs intensify selection and truncate the lower part of the firm distribution.

- Survival declines monotonically as overhead costs rise.
- The figure explains why overhead labor is essential for matching US moments, but also why matching all moments exactly is difficult.

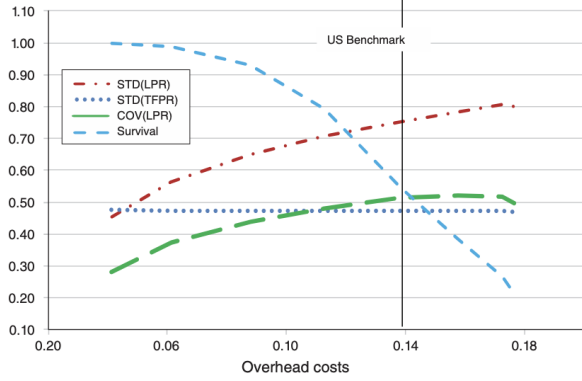


FIGURE 1. THE IMPACT OF OVERHEAD COSTS ON KEY MOMENTS

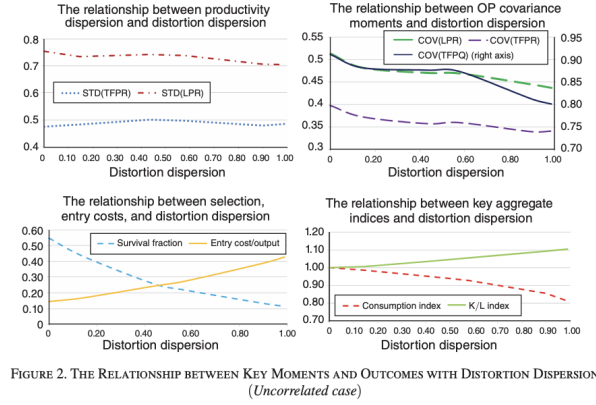


FIGURE 2. THE RELATIONSHIP BETWEEN KEY MOMENTS AND OUTCOMES WITH DISTORTION DISPERSION (Uncorrelated case)

5.4 Figure 2: Distortion dispersion in the uncorrelated case

Read:

- Uncorrelated distortions reduce the OP covariance terms for LPR, TFPR, and TFPQ.
- TFPR dispersion rises only modestly because stronger selection truncates the lower tail.
- Survival falls, the entry-cost share rises, the capital-labor ratio rises, and consumption falls.
- The picture is that distortions worsen allocation, but selection partially masks their effect on dispersion moments.

5.5 Figure 3: Distortion dispersion in the positively correlated case

Read:

- When distortions are positively correlated with productivity, the OP covariance falls much more sharply than in figure 2.
- TFPR dispersion increases strongly; LPR dispersion behaves nonmonotonically.
- Consumption still falls, and the implied welfare effects are large.
- This is the case that best matches countries where productive firms appear unable or unwilling to grow, for example because they are “taxed” more heavily by policy or institutions.

5.6 Table 3: Matching the covariance moments country by country

Read:

- The model is forced to match the OP covariance exactly for every country.

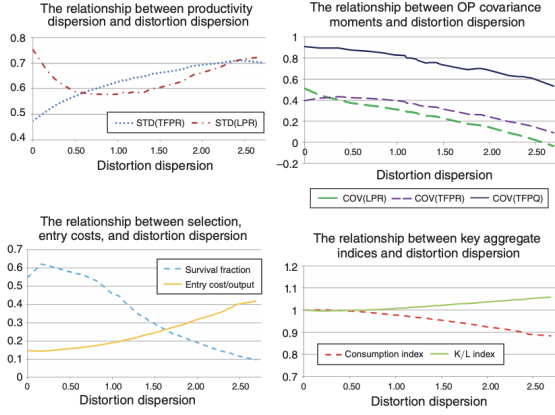


FIGURE 3. THE RELATIONSHIP BETWEEN KEY MOMENTS AND OUTCOMES WITH DISTORTION DISPERSION (Positive correlation case)

TABLE 3—KEY DATA MOMENTS AND MODEL MOMENTS WHEN THE MODEL IS CALIBRATED TO MATCH THE OP LPR COVARIANCE

Country	COV_LPR (Data)	COV_LPR (Model)	STD_LPR (Data)	STD_LPR (Model)	STD_TFPR (Data)	STD_TFPR (Model)	Consumption Index (Model)
United States	0.51	0.51	0.58	0.75	0.39	0.47	1.00
United Kingdom	0.15	0.15	0.59	0.66	0.42	0.69	0.93
Germany	0.28	0.28	0.71	0.59	NA	0.64	0.97
France	0.24	0.24	0.53	0.60	0.23	0.66	0.96
Netherlands	0.30	0.30	0.55	0.59	0.15	0.63	0.97
Hungary	0.16	0.16	1.04	0.65	0.92	0.69	0.93
Romania	-0.03	-0.03	1.05	0.72	0.55	0.70	0.88
Slovenia	0.04	0.04	0.80	0.70	0.22	0.70	0.89

- Once that is done, the implied dispersion moments fit only roughly, especially in transition economies where observed labor-productivity dispersion is very high.
- The implied consumption indices are economically meaningful: around 0.95 for much of Western Europe and closer to 0.88–0.89 for Romania and Slovenia.
- This table is the paper’s strongest quantitative statement that weaker size-productivity relationships are associated with lower aggregate performance.

5.7 Figure 4: Low-overhead robustness

Read:

- With low overhead labor, the model cannot simultaneously match the US covariance moment and the fact that labor-productivity dispersion exceeds TFPR dispersion.
- Distortion dispersion still lowers the covariance between size and productivity, especially for TFPQ.
- But with little overhead labor, some covariance patterns for LPR become more complicated.
- So overhead labor is not a cosmetic modeling choice; it is a central ingredient for fitting the data.

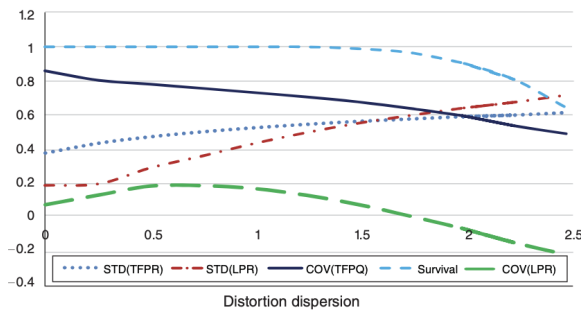


FIGURE 4. THE IMPACT OF DISTORTION DISPERSION ON KEY MOMENTS (Low overhead case)

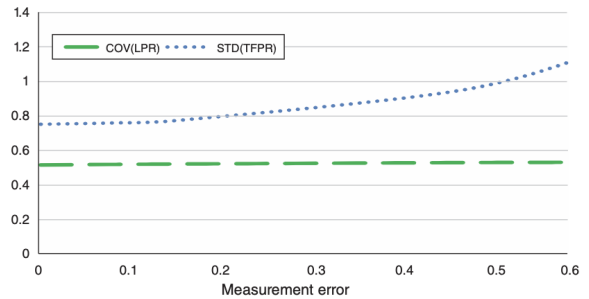


FIGURE 5. THE IMPACT OF MEASUREMENT ERROR ON DISPERSION OF TFPR AND COVARIANCE (LPR)

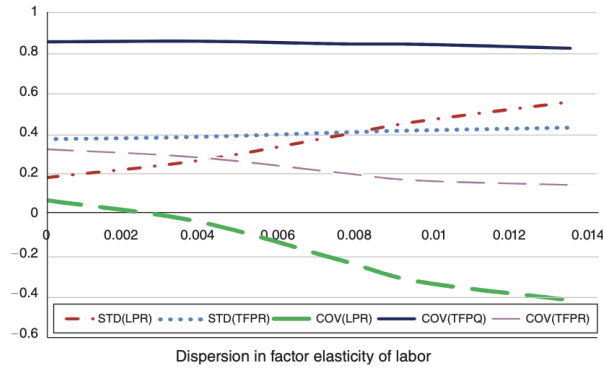


FIGURE 6. THE IMPACT OF DISPERSION IN FACTOR ELASTICITY OF LABOR ON KEY MOMENTS

5.8 Figure 5: Measurement error

Read:

- Revenue measurement error inflates TFPR dispersion substantially.
- The OP covariance based on labor productivity is much less sensitive to classical multiplicative revenue measurement error.
- This supports the paper's claim that covariance moments are empirically more robust than dispersion moments.

5.9 Figure 6: Dispersion in the labor elasticity

Read:

- Allowing dispersion in the labor elasticity can raise labor-productivity dispersion.
- But it pushes the OP covariance in the wrong direction: the covariance falls rather than rises to the US benchmark level.
- So heterogeneity in factor elasticities is not a good substitute for overhead labor as an explanation of the data.

6 Short summary

- **Conclusion.** The paper argues that the covariance between size and productivity is the most informative cross-country moment for studying misallocation.
- **Main identification idea.** Distortions alter both how resources are allocated across surviving firms and which firms survive. A model with overhead labor, quasi-fixed capital, and endogenous entry/exit is needed to map those distortions into the observed moments.
- **Main empirical lesson.** The United States has a much stronger size-productivity relationship than Western Europe or transition economies, and improvements in that covariance during transition are associated with large gains in implied consumption.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that cross-country misallocation is visible not just in productivity dispersion but, more sharply, in the covariance between size and productivity.
- It shows that a heterogeneous-firm model with frictions and endogenous selection can rationalize why labor-productivity dispersion exceeds TFPR dispersion even without distortions.
- It further shows that varying the distortion distribution can reproduce the observed cross-country covariance moments and implies sizable aggregate losses in consumption and GDP per capita.

7.2 Economic intuition

- In a well-functioning economy, more productive firms should be larger. That is the baseline implication of firm heterogeneity models.
- Distortions break this link in two ways. They misallocate resources among incumbents, and they distort which firms survive.
- Overhead labor and quasi-fixed capital matter because they create realistic dispersion in labor productivity even without distortions, so that the paper does not mistakenly attribute every observed productivity gap to policy wedges.
- Once productive firms are prevented from being large, the OP covariance falls, and that decline is closely associated with lower aggregate consumption.

7.3 Takeaway

This paper is best read as a bridge between the misallocation literature and the older Olley–Pakes insight about allocative efficiency. Its key message is that the *size-productivity relationship* is a powerful diagnostic statistic: it is theoretically central, empirically systematic, and more robust than dispersion moments. The model is not rich enough to explain every dispersion pattern in the data, and the authors are explicit about that. But it succeeds in showing that once distortions weaken the size-productivity covariance, the implied aggregate costs are large. For work on productivity, misallocation, transition, or cross-country firm dynamics, this is a foundational reference point.